

03 practice examples on where, having, group by, order by

```
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE Orders_db;
Query OK, 1 row affected (0.10 sec)

mysql> USE Orders_db;
Database changed
mysql> CREATE TABLE orders (
  ->   ord_no INT PRIMARY KEY,
  ->   purch_amt DECIMAL(10,2),
  ->   ord_date DATE,
  ->   customer_id INT,
  ->   salesman_id INT
  -> );
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> INSERT INTO orders VALUES
  -> (70001,150.50,'2012-10-05',3005,5002),
  -> (70009,270.65,'2012-09-10',3001,5005),
  -> (70002,65.26,'2012-10-05',3002,5001),
  -> (70004,110.50,'2012-08-17',3009,5003),
  -> (70007,948.50,'2012-09-10',3005,5002),
  -> (70005,2400.60,'2012-07-27',3007,5001),
  -> (70008,5760.00,'2012-09-10',3002,5001),
  -> (70010,1983.43,'2012-10-10',3004,5006),
  -> (70003,2480.40,'2012-10-10',3009,5003),
  -> (70012,250.45,'2012-06-27',3008,5002),
  -> (70011,75.29,'2012-08-17',3003,5007),
  -> (70013,3045.60,'2012-04-25',3002,5001);
Query OK, 12 rows affected (0.02 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version 5.5.24
t syntax to use near '* FROM orders' at line 1
mysql> SELECT * FROM orders;
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70001	150.50	2012-10-05	3005	5002
70002	65.26	2012-10-05	3002	5001
70003	2480.40	2012-10-10	3009	5003
70004	110.50	2012-08-17	3009	5003
70005	2400.60	2012-07-27	3007	5001
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
70010	1983.43	2012-10-10	3004	5006
70011	75.29	2012-08-17	3003	5007
70012	250.45	2012-06-27	3008	5002
70013	3045.60	2012-04-25	3002	5001

12 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM orders
-> WHERE ord_date='2012-09-10';
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005

3 rows in set (0.01 sec)

```
mysql> SELECT *
-> FROM orders
-> WHERE salesman_id=5001;
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70002	65.26	2012-10-05	3002	5001
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001

4 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM orders
-> ORDER BY purch_amt DESC;
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70010	1983.43	2012-10-10	3004	5006
70007	948.50	2012-09-10	3005	5002
70009	270.65	2012-09-10	3001	5005
70012	250.45	2012-06-27	3008	5002
70001	150.50	2012-10-05	3005	5002
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70002	65.26	2012-10-05	3002	5001

```
mysql>
mysql> SELECT *
  -> FROM orders
  -> ORDER BY ord_date;
```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70013	3045.60	2012-04-25	3002	5001
70012	250.45	2012-06-27	3008	5002
70005	2400.60	2012-07-27	3007	5001
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
70001	150.50	2012-10-05	3005	5002
70002	65.26	2012-10-05	3002	5001
70003	2480.40	2012-10-10	3009	5003
70010	1983.43	2012-10-10	3004	5006

```
12 rows in set (0.00 sec)

mysql> SELECT SUM(purch_amt) AS total_revenue
  -> FROM orders;
```

total_revenue
17541.18

```
1 row in set (0.00 sec)
```

```
mysql> SELECT AVG(purch_amt) AS average_order
  -> FROM orders;
```

average_order
1461.765000

```
1 row in set (0.00 sec)

mysql> SELECT MAX(purch_amt) AS highest_order
  -> FROM orders;
```

highest_order
5760.00

```
1 row in set (0.00 sec)

mysql> SELECT MIN(purch_amt) AS lowest_order
  -> FROM orders;
```

lowest_order
65.26

```
1 row in set (0.00 sec)
```

```
mysql>
mysql> SELECT COUNT(*) AS total_orders
      -> FROM orders;
+-----+
| total_orders |
+-----+
|          12 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT salesman_id,
      ->      SUM(purch_amt) AS total_sales
      -> FROM orders
      -> GROUP BY salesman_id;
+-----+-----+
| salesman_id | total_sales |
+-----+-----+
|          5002 |      1349.45 |
|          5001 |     11271.46 |
|          5003 |      2590.90 |
|          5005 |       270.65 |
|          5006 |      1983.43 |
|          5007 |        75.29 |
+-----+-----+
6 rows in set (0.01 sec)
```

```
mysql> SELECT customer_id,
      ->      SUM(purch_amt) AS total_purchase
      -> FROM orders
      -> GROUP BY customer_id;
+-----+-----+
| customer_id | total_purchase |
+-----+-----+
|          3005 |      1099.00 |
|          3002 |     8870.86 |
|          3009 |      2590.90 |
|          3007 |      2400.60 |
|          3001 |       270.65 |
|          3004 |      1983.43 |
|          3003 |        75.29 |
|          3008 |       250.45 |
+-----+-----+
8 rows in set (0.00 sec)
```

```
mysql> SELECT customer_id,
      ->      SUM(purch_amt) AS total_purchase
      -> FROM orders
      -> GROUP BY customer_id;
+-----+-----+
| customer_id | total_purchase |
+-----+-----+
|          3005 |      1099.00 |
|          3002 |     8870.86 |
|          3009 |      2590.90 |
|          3007 |      2400.60 |
|          3001 |       270.65 |
|          3004 |      1983.43 |
|          3003 |        75.29 |
|          3008 |       250.45 |
+-----+-----+
8 rows in set (0.00 sec)
```

```
mysql> SELECT customer_id,
->         MAX(purch_amt) AS highest_purchase
-> FROM orders
-> GROUP BY customer_id;
```

customer_id	highest_purchase
3005	948.50
3002	5760.00
3009	2480.40
3007	2400.60
3001	270.65
3004	1983.43
3003	75.29
3008	250.45

8 rows in set (0.00 sec)

```
mysql>
mysql> SELECT salesman_id,
->         SUM(purch_amt) AS total_sales
-> FROM orders
-> GROUP BY salesman_id
-> HAVING SUM(purch_amt) > 3000;
```

salesman_id	total_sales
5001	11271.46

1 row in set (0.00 sec)

```
mysql> SELECT customer_id,
->         SUM(purch_amt) AS total_purchase
-> FROM orders
-> GROUP BY customer_id
-> HAVING SUM(purch_amt) > 2500;
```

customer_id	total_purchase
3002	8870.86
3009	2590.90

2 rows in set (0.00 sec)

```
mysql>
mysql> SELECT customer_id,
->         COUNT(*) AS total_orders
-> FROM orders
-> GROUP BY customer_id
-> HAVING COUNT(*) > 1;
```

customer_id	total_orders
3005	2
3002	3
3009	2

3 rows in set (0.00 sec)

```
mysql> SELECT customer_id,
->         SUM(purch_amt) AS total_purchase
-> FROM orders
-> GROUP BY customer_id
-> HAVING SUM(purch_amt) > 1000
```

```
mysql> SELECT customer_id,
->      SUM(purch_amt) AS total_purchase
-> FROM orders
-> GROUP BY customer_id
-> HAVING SUM(purch_amt) > 1000
-> ORDER BY total_purchase DESC;
```

customer_id	total_purchase
3002	8870.86
3009	2590.90
3007	2400.60
3004	1983.43
3005	1099.00

5 rows in set (0.00 sec)

```
mysql> SELECT customer_id,
->      MAX(purch_amt) AS max_purchase
-> FROM orders
-> GROUP BY customer_id
-> HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
```

customer_id	max_purchase
3002	5760.00
3009	2480.40
3007	2400.60

3 rows in set (0.00 sec)

```
mysql> SELECT salesman_id,
->      COUNT(*) AS total_orders
-> FROM orders
-> GROUP BY salesman_id
-> HAVING COUNT(*) >= 2;
```

salesman_id	total_orders
5002	3
5001	4
5003	2

3 rows in set (0.00 sec)

```
mysql> SELECT ord_date,
->      MAX(purch_amt) AS highest_purchase
-> FROM orders
-> GROUP BY ord_date
-> HAVING MAX(purch_amt) > 2000;
```

ord_date	highest_purchase
2012-10-10	2480.40
2012-07-27	2400.60
2012-09-10	5760.00
2012-04-25	3045.60

4 rows in set (0.00 sec)